28 An electron and a proton are both accelerated from rest through the same electric potential difference.

Which row correctly compares the kinetic energy and the de Broglie wavelength of the electron and proton after the acceleration?

	kinetic energy	de Broglie wavelength
A	electron has greater energy	proton has longer wavelength
B	proton has greater energy	electron has longer wavelength
C	both have the same energy	proton has longer wavelength
D	both have the same energy	electron has longer wavelength